PONTSHO MBIZO

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SUMMARY

Final-year Electrical & Computer Engineering student with four industry placements across defence, mining, and infrastructure. I build hardware and embedded systems that work and check them against measurements: a weeding gantry robot with closed-loop control and on-device vision, browser-based FPGA, microcontroller and control-system simulators, and parameterised tools that engineering teams have adopted. I communicate technical findings clearly to diverse stakeholders.

PROJECTS

Weeding Gantry Robot — final-year research project (EEE4022S) *2026*

- Built a low-cost three-axis gantry (Raspberry Pi Zero W, ESP32-C3) that photographs a crop bed, classifies weeds on the device and sprays each one; wrote the Python operator software and the ESP32 closed-loop motion firmware with homing, stall detection and a safety watchdog.
- 245 closed-loop moves ended within one sensor count (1.07 mm); worst settling time 1.86 s against a 3 s limit; positioning within 14 mm against a 20 mm requirement. Spray-grid detection F1 0.908 for segmentation against 0.654 for YOLOv8n.

fpgasim and mcusim — in-browser hardware simulators *2025 – 2026*

- **fpgasim:** Verilog in the browser, simulated with Icarus Verilog in ephemeral, network-less containers, on a Tang Nano 9K board model with waveforms; 69 unit tests, 272–375 ms round trip.
- **mcusim:** bare-metal C for an STM32F103 Blue Pill, compiled with arm-none-eabi-gcc and run in QEMU, with the GPIO trace drawn on a board view; 68 unit tests, 357–469 ms round trip.

scope.io — interview-prep platform for electronics engineers *2026*

- LeetCode-style platform (Next.js, Firebase) that grades circuit, FPGA and firmware solutions on the server against reference waveforms; 26 problems written, 126 unit tests.

fulgurlab — Simulink-style block-diagram simulator *2026*

- React editor and Node server that compiles diagrams to Xcos models and simulates them with headless Scilab in a hardened container; 19 block types checked against analytic answers, about 0.23 s per run.

SRAM data capture — MAX10 FPGA (published from the Cybicom internship) *2025 – 2026*

- Found and fixed a timing bug in my own capture logic while publishing it: re-armed the timer per access and synchronised the enable lines; a phase-sweep testbench went from 2 to 16 of 16 correct captures in simulation.

WORK EXPERIENCE

Project Engineering Intern, Tharisa Minerals *Dec 2025 – Jan 2026*

- Built a parameterised three-phase MCC sizing tool in Excel: full-load currents, voltage drop, power factor, and operating cost per motor — validated analytically and against manufacturer data (SANS 1507, WEG IE3 motors).
- Built-in percentage-deviation check surfaces specification errors before procurement; methodology report accepted by engineering department.
- Tool adopted by the site team as the baseline for all feeder studies; presented cost-comparison results to the project engineering team.

Engineering Intern, Cybicom Atlas Defence *Jun – Jul 2025*

- Designed a high-speed FPGA capture system (Verilog, Intel MAX10) to non-destructively read a battery-backed SRAM on a legacy control system that presents data for about 10 ns, 70 ns after enable — 200 MHz PLL, 5 ns timer, 5 V to 3.3 V buffer stage.
- Built a bidirectional, multi-threaded UDP communication system in C (POSIX threads) for a motion simulator, with custom checksums and full handover documentation.
- Designed and 3D-printed a parametric mounting system in Fusion 360 to secure a 4U simulation computer against inertial loads inside a simulator instructor station.
- Participated in an Acceptance Test Procedure for a full motion simulation system across mechanical, electrical, and software subsystems.

Network Engineer/Artisan (Vacation Work), Tharisa Minerals *Dec 2024 – Jan 2025*

- Restored LAN connectivity across multiple plant sites; investigated Microsoft Cloud PC as a cost-reduction strategy for remote operations.
- Expanded network infrastructure by connecting new office spaces and improving plant-wide network access.

Electrical and Instrumentation Vacation Work, Tharisa Minerals *Dec 2023 – Jan 2024*

- Assisted Electrical Artisans and Instrumentation Technicians in routine maintenance, inspection, and upkeep of electrical equipment and control systems.

EDUCATION

BSc Eng Electrical and Computer Engineering (ECSA-accredited) *2023 – Present*

University of Cape Town

- Final-year: High Performance Embedded Systems, DSP & ML, Controls & Instrumentation, Law for Engineers; Design project — oceanographic sensor buoy (full systems engineering lifecycle).
- Earlier coursework: Embedded Systems, Electronic Devices & Circuits, Control Systems, Programming.

ADDITIONAL INFORMATION

- **Software Development:** C/C++, Java, Python (NumPy, pandas, scikit-learn, OpenCV), JavaScript/TypeScript (React, Next.js, Node.js), MATLAB, Scilab/Xcos
- **Hardware & Firmware:** Verilog HDL, FPGA (Intel MAX10, Gowin), STM32, ESP32, Raspberry Pi, QEMU, Icarus Verilog, KiCad, Intel Quartus Prime
- **Systems & Tools:** Docker/Podman, Firebase, Git and GitHub Actions, Linux (Fedora), ROS 2, Fusion 360, Onshape & FeatureScript, LaTeX, MS Excel
- **Languages:** English (Fluent), Setswana (Native), isiZulu (Basic), isiXhosa (Basic)
- **Driver's License:** None
- **Awards / Activities:**
  1. UCT Plus Silver — 45 Hours Role Modelling Leadership Award for residence mentoring.
  2. Dell Young Leader.
  3. Electrical Engineering peer mentor at UCT.

REFERENCES

Jason Suter, Systems Engineer
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Robey Labuschagne, Project Engineer
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